

EX. NO: 4

IMPLEMENTATION OF VIGENÈRE CIPHER

AIM

To write a C program to implement the Vigenère Cipher substitution technique for encrypting plaintext using a keyword.

ALGORITHM

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the keyword from the user.

Step 4: Convert both the plaintext and keyword to uppercase.

Step 5: Repeat the keyword until its length matches the length of the plaintext.

Step 6: Encrypt each character using the Vigenère Cipher formula:

$$C_i = (P_i + K_i) \text{ } 26$$

where:

- P = Plaintext letter
- K = Keyword letter
- C = Ciphertext letter

Step 7: Convert the encrypted values back to alphabetic characters.

Step 8: Display the ciphertext.

Step 9: Stop the program.

PROGRAM:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main()
{
    char plain[100], key[100], newKey[100], cipher[100];
    int i, j = 0, len;
    printf("Enter Plain Text: ");
    scanf("%s", plain);
    printf("Enter Keyword: ");
    scanf("%s", key);
```

```

len = strlen(plain);
// Convert to uppercase
for(i = 0; i < len; i++)
    plain[i] = toupper(plain[i]);
for(i = 0; key[i] != '\0'; i++)
    key[i] = toupper(key[i]);
// Generate repeated key
for(i = 0; i < len; i++)
{
    newKey[i] = key[j];
    j++;

    if(key[j] == '\0')
        j = 0;
}
newKey[len] = '\0';
// Encryption
for(i = 0; i < len; i++)
{
    cipher[i] = ((plain[i] - 'A') + (newKey[i] - 'A')) % 26 + 'A';
}
cipher[len] = '\0';
printf("\nGenerated Key : %s", newKey);
printf("\nCipher Text  : %s\n", cipher);
return 0;
}

```

OUTPUT:

The screenshot displays the Programiz C Online Compiler interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The compiler window is titled "main.c" and contains the following C code:

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main()
6 {
7     char plain[100], key[100], newKey[100], cipher[100];
8     int i, j = 0, len;
9
10    printf("Enter Plain Text: ");
11    scanf("%s", plain);
12
13    printf("Enter Keyword: ");
14    scanf("%s", key);
15
16    len = strlen(plain);
17
18    // Convert to uppercase
19    for(i = 0; i < len; i++)
20        plain[i] = toupper(plain[i]);
21
22    for(i = 0; key[i] != '\0'; i++)
23        key[i] = toupper(key[i]);
24}
```

The "Output" panel on the right shows the execution results:

```
Enter Plain Text: pallavi
Enter Keyword: crypto
Generated Key : CRYPTOC
Cipher Text : RRJATJK

=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the system clock indicating 09:05 PM on 03-08-2026, and the weather as 28°C with light rain.